

Quantitative Pathway for Coupling Atmosphere, Hydrosphere, Interior Thermal State, and Geochemistry

LIFEORIG working note

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1 Purpose

This document defines a quantitative pathway for deciding whether a planet has a hydrosphere and, if it does, how the hydrosphere should provide boundary conditions for ocean and geochemical calculations.

The intended sequence is:

1. Solve the atmosphere.
2. Use the atmosphere plus planet structure to evaluate liquid/ice water.
3. Decide whether there is no hydrosphere, a surface ocean/pond, or an ice shell plus subsurface ocean.
4. Use the hydrosphere state to define the pressure, temperature, composition, and mineral boundary conditions for PHREEQC/geochemistry.

This note is written to be implementable. The details of specific ocean/ice layer models can replace the simple formulas below, but the data flow should remain the same.

2 Atmospheric Inputs

After the atmospheric solver converges, the hydrosphere module should receive:

$$z_i \quad \text{altitude grid [m]}, \tag{1}$$

$$T_i \quad \text{temperature profile [K]}, \tag{2}$$

$$P_i \quad \text{pressure profile [Pa]}, \tag{3}$$

$$n_{\text{H}_2\text{O},i} \quad \text{H}_2\text{O number density profile [m}^{-3}\text{]}, \tag{4}$$

$$T_s = T(z = 0), \tag{5}$$

$$P_s = P(z = 0). \tag{6}$$

The atmosphere is the source of surface boundary conditions and of the condensable vapor reservoir. If outgassing, delivery, or non-equilibrium chemistry adds water to the system, it enters the atmosphere first and is then seen by the hydrosphere through $n_{\text{H}_2\text{O}}(z)$.

3 Planetary Inputs

The planet model should provide:

$$M_p \quad \text{planet mass [kg]}, \quad (7)$$

$$R_p \quad \text{planet radius [m]}, \quad (8)$$

$$g_s = \frac{GM_p}{R_p^2}, \quad (9)$$

$$\rho_p = \frac{M_p}{(4/3)\pi R_p^3}. \quad (10)$$

The bulk density ρ_p is not enough to determine composition uniquely, but it can provide a first-order estimate of rock/ice fraction. This is especially important for Europa-like bodies, where the ocean is not produced by atmospheric condensation.

4 Surface Liquid Water from Atmospheric Condensation

4.1 Saturation vapor pressure

For each atmospheric layer, compute the saturation vapor pressure of water:

$$p_{\text{sat}}(T_i). \quad (11)$$

A first implementation can use the Magnus formula:

$$p_{\text{sat}}(T) = 610.94 \exp \left[\frac{17.625 T_C}{T_C + 243.04} \right] \quad [\text{Pa}], \quad (12)$$

where

$$T_C = T[\text{K}] - 273.15. \quad (13)$$

This is a practical approximation near ordinary liquid-water conditions. More accurate vapor pressure formulas can replace it later.

4.2 Saturation number density

The saturation number density of water vapor is

$$n_{\text{sat},i} = \frac{p_{\text{sat}}(T_i)}{k_B T_i}. \quad (14)$$

The excess water vapor number density is

$$n_{\text{excess},i} = \max[0, n_{\text{H}_2\text{O},i} - n_{\text{sat},i}]. \quad (15)$$

4.3 Condensed water mass

The atmospheric water vapor mass is

$$M_{\text{H}_2\text{O},\text{atm}} = \int_0^{z_{\text{top}}} n_{\text{H}_2\text{O}}(z) m_{\text{H}_2\text{O}} 4\pi (R_p + z)^2 dz. \quad (16)$$

The saturation-capacity water mass is

$$M_{\text{H}_2\text{O},\text{sat}} = \int_0^{z_{\text{top}}} n_{\text{sat}}(z) m_{\text{H}_2\text{O}} 4\pi(R_p + z)^2 dz. \quad (17)$$

The condensed water mass implied by the atmospheric solution is

$$M_{\text{cond}} = \int_0^{z_{\text{top}}} n_{\text{excess}}(z) m_{\text{H}_2\text{O}} 4\pi(R_p + z)^2 dz. \quad (18)$$

This is the surface-liquid reservoir directly implied by the atmosphere.

4.4 Surface liquid condition

Surface liquid water is allowed only if:

$$P_s > P_{\text{triple}} = 611.657 \text{ Pa}, \quad (19)$$

$$T_s \geq 273.15 \text{ K}, \quad (20)$$

$$T_s \leq T_{\text{boil}}(P_s), \quad (21)$$

$$M_{\text{cond}} > 0. \quad (22)$$

If these conditions are satisfied, the equivalent global surface-ocean depth is

$$d_{\text{surf}} = \frac{M_{\text{cond}}}{4\pi R_p^2 \rho_w}, \quad (23)$$

where $\rho_w \approx 1000 \text{ kg m}^{-3}$.

The bottom pressure of that surface ocean is

$$P_{\text{bottom,surf}} = P_s + \rho_w g_s d_{\text{surf}}. \quad (24)$$

5 Interior Water/Ice Estimate from Bulk Density

Atmospheric condensation cannot produce a Europa-like subsurface ocean. For icy bodies, estimate a bulk ice/water fraction from the planet density.

5.1 Two-component rock/ice model

Assume a first-order two-component body:

$$\rho_p = x_{\text{ice}} \rho_{\text{ice}} + (1 - x_{\text{ice}}) \rho_{\text{rock}}, \quad (25)$$

where x_{ice} is a volume fraction.

Solving:

$$x_{\text{ice}} = \frac{\rho_{\text{rock}} - \rho_p}{\rho_{\text{rock}} - \rho_{\text{ice}}}. \quad (26)$$

In implementation:

$$x_{\text{ice}} \leftarrow \text{clip}(x_{\text{ice}}, 0, 1). \quad (27)$$

Use first-pass constants:

$$\rho_{\text{ice}} \approx 1000 \text{ kg m}^{-3}, \quad (28)$$

$$\rho_{\text{rock}} \approx 3300 \text{ kg m}^{-3}. \quad (29)$$

The corresponding mass fraction is

$$f_{\text{ice, mass}} = \frac{x_{\text{ice}} \rho_{\text{ice}}}{\rho_p}. \quad (30)$$

The total ice/water mass estimate is

$$M_{\text{ice/water}} = f_{\text{ice, mass}} M_p. \quad (31)$$

The equivalent global water-layer thickness is

$$d_{\text{water, eq}} = \frac{M_{\text{ice/water}}}{4\pi R_p^2 \rho_w}. \quad (32)$$

5.2 Interpretation

If $\rho_p > \rho_{\text{rock}}$, the two-component model gives no ice/water component. The body is treated as rocky/metal-rich in this first-pass model.

If $\rho_p < \rho_{\text{ice}}$, the two-component model is insufficient: porosity, low-density volatiles, or gas-rich structure may be required.

For Europa-like bodies, this model gives a physically motivated water/ice reservoir that does not come from atmospheric vapor.

6 Thermal Model for Ice Shell and Subsurface Ocean

6.1 Interior heat flux

The interior model should provide an upward heat flux:

$$q_{\text{int}} \quad [\text{W m}^{-2}]. \quad (33)$$

This can come from:

- radiogenic heating,
- tidal heating,
- secular cooling,
- parameterized convection in the rocky mantle,
- a user-specified first-pass value.

6.2 Conductive ice shell thickness

For an icy body with surface temperature below melting, a first conductive ice shell estimate is:

$$q_{\text{int}} = k_{\text{ice}} \frac{T_m - T_s}{d_{\text{ice}}}, \quad (34)$$

so

$$d_{\text{ice}} = k_{\text{ice}} \frac{T_m - T_s}{q_{\text{int}}}. \quad (35)$$

Use first-pass constants:

$$k_{\text{ice}} \approx 2.5 \text{ W m}^{-1} \text{ K}^{-1}, \quad (36)$$

$$T_m \approx 273.15 \text{ K}. \quad (37)$$

The model should allow replacement of T_m by a pressure- and salinity-dependent melting curve.

6.3 Subsurface ocean condition

A subsurface ocean exists in the first-pass model if:

$$M_{\text{ice/water}} > 0, \quad (38)$$

$$T_s < T_m, \quad (39)$$

$$q_{\text{int}} > 0, \quad (40)$$

$$d_{\text{water,eq}} > d_{\text{ice}}. \quad (41)$$

Then:

$$d_{\text{ocean,sub}} = d_{\text{water,eq}} - d_{\text{ice}}, \quad (42)$$

$$M_{\text{ocean,sub}} = 4\pi R_p^2 \rho_w d_{\text{ocean,sub}}. \quad (43)$$

The top pressure of the subsurface ocean is

$$P_{\text{ocean,top}} = P_s + \rho_{\text{ice}} g_s d_{\text{ice}}. \quad (44)$$

The bottom pressure is

$$P_{\text{ocean,bottom}} = P_{\text{ocean,top}} + \rho_w g_s d_{\text{ocean,sub}}. \quad (45)$$

7 Ice Shell Convection Check

The conductive shell estimate should be followed by a convection check. A simple Rayleigh number is:

$$Ra_{\text{ice}} = \frac{\rho_{\text{ice}} g_s \alpha_{\text{ice}} \Delta T d_{\text{ice}}^3}{\kappa_{\text{ice}} \eta_{\text{ice}}}, \quad (46)$$

where

$$\Delta T = T_m - T_s, \quad (47)$$

$$\alpha_{\text{ice}} \quad \text{thermal expansivity}, \quad (48)$$

$$\kappa_{\text{ice}} \quad \text{thermal diffusivity}, \quad (49)$$

$$\eta_{\text{ice}} \quad \text{effective ice viscosity}. \quad (50)$$

If

$$Ra_{\text{ice}} > Ra_{\text{crit}}, \quad (51)$$

then a purely conductive shell is not self-consistent. A first critical value is

$$Ra_{\text{crit}} \sim 10^3, \quad (52)$$

but stagnant-lid ice convection uses more detailed scalings.

In code, this can initially be stored as a diagnostic:

$$\text{ice_shell_regime} = \begin{cases} \text{conductive}, & Ra_{\text{ice}} \leq Ra_{\text{crit}}, \\ \text{convective}, & Ra_{\text{ice}} > Ra_{\text{crit}}. \end{cases} \quad (53)$$

8 Rocky Interior Thermal Evolution

For rocky planets or rocky cores, use parameterized convection rather than full 2D/3D convection as the first implementation.

8.1 Energy balance

Let T_m be a representative mantle/core temperature. A simple thermal evolution equation is:

$$\rho_m C_p V_m \frac{dT_m}{dt} = H(t) - Q_{\text{out}}, \quad (54)$$

where:

$$\rho_m \quad \text{mantle/core density,} \quad (55)$$

$$C_p \quad \text{heat capacity,} \quad (56)$$

$$V_m \quad \text{convecting volume,} \quad (57)$$

$$H(t) \quad \text{internal heat production,} \quad (58)$$

$$Q_{\text{out}} \quad \text{heat loss.} \quad (59)$$

The heat loss is also the lower-boundary heat source for the atmosphere and surface energy balance. If Q_{out} is a total power, then the surface internal flux is:

$$F_{\text{int}} = \frac{Q_{\text{out}}}{4\pi R_p^2}. \quad (60)$$

If the interior model directly computes a heat flux q_{int} , then:

$$F_{\text{int}} = q_{\text{int}}. \quad (61)$$

8.2 Internal heat production

The total internal heat production is decomposed as:

$$H(t) = H_{\text{rad}}(t) + H_{\text{tidal}}(t) + H_{\text{grav}}(t). \quad (62)$$

8.2.1 Radiogenic heating

For a detailed isotope model:

$$H_{\text{rad}}(t) = \sum_i H_{i,0} \exp(-\lambda_i t), \quad (63)$$

where

$$\lambda_i = \frac{\ln 2}{t_{1/2,i}}. \quad (64)$$

The main long-lived isotopes in rocky interiors are:

$$^{238}\text{U}, \quad ^{235}\text{U}, \quad ^{232}\text{Th}, \quad ^{40}\text{K}. \quad (65)$$

For a first implementation, a lumped effective decay law is acceptable:

$$H_{\text{rad}}(t) = H_{\text{rad},0} \exp\left(-\frac{t}{\tau_{\text{rad}}}\right). \quad (66)$$

8.2.2 Tidal heating

For a synchronously rotating satellite or close-in planet with eccentricity tides, a first-order total tidal power is:

$$H_{\text{tidal}} = \frac{21}{2} \frac{k_2}{Q} \frac{GM_\star^2 R_p^5 n e^2}{a^6}, \quad (67)$$

where:

$$k_2 \quad \text{second Love number,} \quad (68)$$

$$Q \quad \text{tidal quality factor,} \quad (69)$$

$$M_\star \quad \text{host/primary mass,} \quad (70)$$

$$R_p \quad \text{planet/moon radius,} \quad (71)$$

$$n \quad \text{mean motion,} \quad (72)$$

$$e \quad \text{orbital eccentricity,} \quad (73)$$

$$a \quad \text{orbital semi – major axis.} \quad (74)$$

The mean motion is:

$$n = \sqrt{\frac{GM_\star}{a^3}}. \quad (75)$$

The corresponding globally averaged tidal heat flux is:

$$F_{\text{tidal}} = \frac{H_{\text{tidal}}}{4\pi R_p^2}. \quad (76)$$

This term is essential for Europa-like cases, where the atmosphere does not control the existence of the subsurface ocean but tidal heat can maintain a liquid layer below an ice shell.

8.2.3 Gravitational and secular heat

Part of gravitational/secular cooling is already represented by the thermal inertia term on the left-hand side of the energy equation:

$$\rho_m C_p V_m \frac{dT_m}{dt}. \quad (77)$$

If an additional early-time gravitational energy source is desired, use a parameterized form:

$$H_{\text{grav}}(t) = H_{\text{grav},0} \exp\left(-\frac{t}{\tau_{\text{grav}}}\right). \quad (78)$$

This can represent early differentiation, contraction, or accretional residual heat in a first-pass model.

8.3 Nusselt-Rayleigh scaling

The convective heat flux can be parameterized as:

$$Nu = a Ra^\beta. \quad (79)$$

Then:

$$q_{\text{conv}} = Nu \frac{k_m (T_m - T_s)}{D_m}, \quad (80)$$

and

$$Q_{\text{out}} = 4\pi R_p^2 q_{\text{conv}}. \quad (81)$$

The Rayleigh number is:

$$Ra = \frac{\rho_m g_s \alpha_m (T_m - T_s) D_m^3}{\kappa_m \eta_m}. \quad (82)$$

First-pass values often use $\beta \approx 1/3$, but stagnant-lid convection requires care because viscosity is strongly temperature dependent.

9 Hydrosphere State Classification

After computing surface condensation and interior ice/water state, classify the planet:

1. No hydrosphere

$$M_{\text{cond}} = 0, \quad (83)$$

$$M_{\text{ice/water}} = 0 \quad (84)$$

or liquid/ice conditions fail.

2. Surface ocean or pond

$$M_{\text{cond}} > 0, \quad (85)$$

$$P_s > P_{\text{triple}}, \quad (86)$$

$$273.15 \text{ K} \leq T_s \leq T_{\text{boil}}(P_s). \quad (87)$$

3. Ice shell with subsurface ocean

$$T_s < T_m, \quad (88)$$

$$d_{\text{water,eq}} > d_{\text{ice}}. \quad (89)$$

4. Ice-only hydrosphere

$$T_s < T_m, \quad (90)$$

$$d_{\text{water,eq}} \leq d_{\text{ice}}. \quad (91)$$

10 Geochemical Boundary Conditions

PHREEQC/geochemistry should run only after the hydrosphere state is known.

10.1 Surface pond or surface ocean

For a surface pond/ocean:

$$T_{\text{geo}} = T_s, \quad (92)$$

$$P_{\text{geo}} = P_s + \rho_w g_s d_{\text{surf}}, \quad (93)$$

$$\text{composition} = \text{planet.hydro}[\text{ocean_composition}], \quad (94)$$

$$\text{minerals} = \text{surface rock preset}. \quad (95)$$

This corresponds to surface aqueous geochemistry, weathering, and volcanic-rock wetting environments.

10.2 Volcanic rock environment

For volcanic-rock chemistry:

$$T_{\text{geo}} = T_s \quad \text{or a volcanic surface temperature,} \quad (96)$$

$$P_{\text{geo}} = P_s, \quad (97)$$

$$\text{fluid} = \text{condensed water/pond composition,} \quad (98)$$

$$\text{minerals} = \text{basaltic/volcanic rock assemblage.} \quad (99)$$

This environment is relevant when surface liquid water exists but does not cover all rocky surface. The model should track available exposed rock fraction if the surface ocean depth is shallow.

10.3 Hydrothermal vent

For hydrothermal vent chemistry:

$$T_{\text{geo}} = T_{\text{reaction depth}}, \quad (100)$$

$$P_{\text{geo}} = P_{\text{ocean, bottom}}, \quad (101)$$

$$\text{fluid} = \text{ocean composition,} \quad (102)$$

$$\text{minerals} = \text{rocky core/basalt/ultramafic assemblage.} \quad (103)$$

The reaction temperature should come from the interior thermal model. A first implementation can choose a reaction depth or a temperature window, for example:

$$T_{\text{reaction}} \in [350, 650] \text{ K.} \quad (104)$$

The hydrothermal flux to the ocean can later be written:

$$F_j = \dot{M}_{\text{fluid}} (C_{j,\text{vent}} - C_{j,\text{ocean}}), \quad (105)$$

where $\dot{M}_{\text{fluid}}$ is a water circulation rate and C_j is the concentration of species j .

11 Suggested Code Interface

The main implementation target should be:

```
compute_hydrosphere_state(  
    planet_mass,  
    planet_radius,  
    surface_temperature,  
    surface_pressure,  
    gravity,  
    altitude,  
    temperature_profile,  
    h2o_number_density,  
    internal_heat_flux,  
    rho_rock=3300 kg/m3,  
    rho_ice=1000 kg/m3,  
)
```

It should return a state object containing:

```
has_surface_ocean
surface_ocean_mass
surface_ocean_depth
has_subsurface_ocean
ice_shell_thickness
subsurface_ocean_mass
subsurface_ocean_depth
ice_shell_regime
ocean_top_pressure
ocean_bottom_pressure
hydrosphere_class
```

12 References

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